| | | |
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| Yanbu Industrial College<br>Computer Science & Engineering Dept.<br>Information & Computer Technology Dept. | | كلية ينبع الصناعية<br>قسم علوم وهندسة الحاسب الآلي<br>قسم تقنية المعلومات والحاسب الآلي |

**PROJECT**

**ACADEMIC YEAR 1445/1446 H (2024/2025 G), SEMESTER II - 462**

# DATABASE SYSTEMS
# CS 311

| | |
|---|---|
| **STUDENT NAME:** | Wafa [redacted]<br>Shaden [redacted]<br>Amjad Mohammed<br>Sarah [redacted] |
| **SECTION:** | 2 |
| **DATE:** | **February 16, 2025** |

| FOR INSTRUCTOR USE ONLY | GENERAL INSTRUCTIONS |
|---|---|

| Q. No. | CLOs | MAX MARK | MARKS OBTAINED |
|---|---|---|---|
| CD1 | 1.03<br>2.02 | 5 | |
| CD2 | 2.02<br>3.01 | 10 | |
| LD1 | 2.02<br>3.01 | 10 | |
| LD2 | 3.3 | 10 | |
| F | 3.02 | 5 | |
| TOTAL MARKS | | 40 | |

- **Write your name and I.D. number in the above space.**
- **Submission is required in soft-copy form (a PDF format - copy-paste codes into this form).**
- **Copied work will not be graded.**

| | |
|---|---|
| **MARKED BY:** | **Signature:** |
| **CHECKED BY:** | **Signature:** |

# Salon Reservation Management System

## 1. Conceptual Design

### Phase I :

## introduction

Our project focuses on designing and building a comprehensive Salon Reservation System that aims to simplify and efficiently manage customer appointments, salon services, and employee assignments. The system is tailored to enhance daily operations within the salon by streamlining the booking process, organizing service categories, and ensuring specialists are assigned effectively based on their expertise.

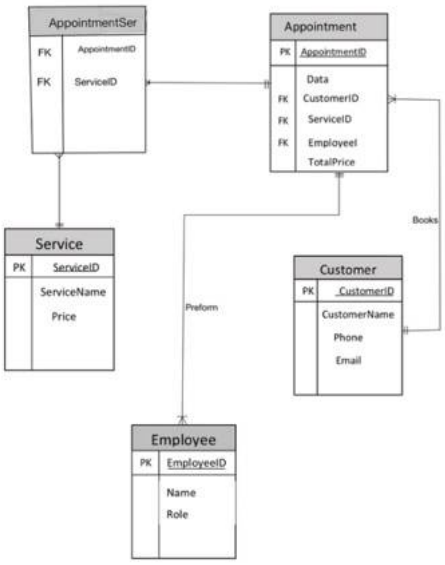
This solution is designed not only to improve internal operations but also to enhance the overall customer experience. Clients will be able to view available services, book appointments during working hours, and choose payment methods securely. On the other hand, employees and managers can monitor schedules, manage service categories, and track transactions — all through a well-structured database system.

## Business Rules

- Customers can book an appointment only during working hours (9:00 AM – 11:00 PM).
- The payment method must be either 'Mada card' or 'Bank Transfer'.
- Each appointment must have a status to indicate whether it is booked, cancelled, or completed.
- Each service must have a duration that specifies how long it takes to complete.
- The manager oversees all service categories and supervises all employees working under these categories

## First draft of ERD

In the first ERD draft, we focused only on creating the core tables such as Customer, Service, Employee, and Appointment. However, at this stage, we did not consider the importance of additional supporting tables like Payment, ServiceCategory, or Manager, which are essential for enforcing business rules and ensuring complete data structure.



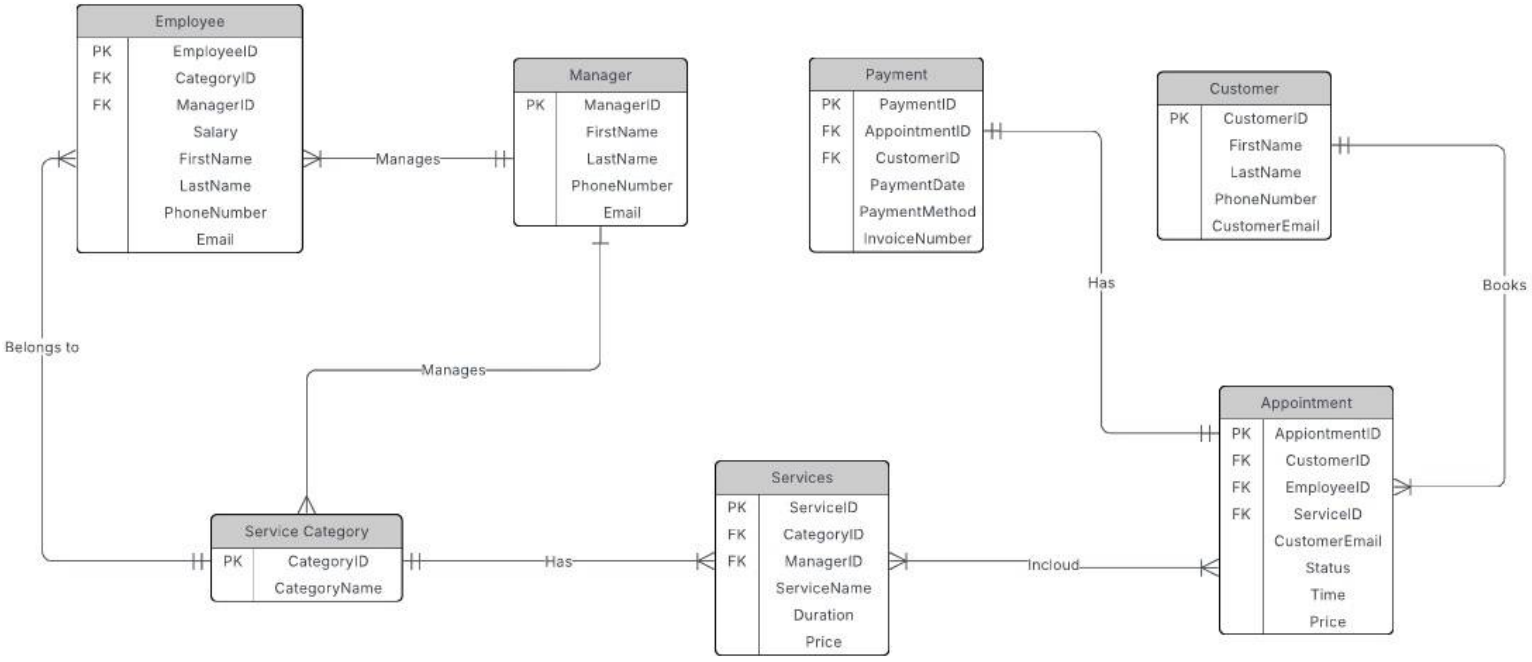
**Phase II :**

## Second draft of ERD

In this second draft, we initially considered removing the Service Category table and placing it as a column in the Employee table. However, this caused a First Normal Form (1NF) violation due to repeated values for the same category across multiple employees. Since 1NF requires atomic values, we kept the category as a separate table.

We also created a Manager table, which we later realized may not be necessary because the number of managers is limited. Nonetheless, it was retained in this version to explore managerial relationships more clearly.

Additionally, we noticed a 1NF issue in the Appointment table, where the ServiceID field held multiple values in one row.



# Entities & Attributes

## Customer Entity & Attributes

Primary key: CustomerID
Foreign key: None

| CustomerID | FirstName | LastName | PhoneNumber | CustomerEmail |
|---|---|---|---|---|
| 1 | Amjad | Mohammed | 0511632582 | amjad.mohammed@gmail.com |
| 2 | Sara | Alhassan | 0552345678 | sara.alhassan@gmail.com |
| 3 | Shaden | Mimesh | 0591863552 | shaden.mimesh@gmail.com |
| 4 | Wafa | Alzahrani | 0531845544 | wafa.alzahrani@gmail.com |
| 5 | Lama | Khalid | 0521023344 | lama.khalid@gmail.com |
| 6 | Hassan | Ali | 0561122334 | hassan.ali@gmail.com |
| 7 | Rania | Fahad | 0509988776 | rania.fahad@gmail.com |
| 8 | Noura | Saad | 0543436789 | noura.saad@gmail.com |
| 9 | Joud | Sami | 0581122333 | joud.sami@gmail.com |
| 10 | Maha | Alqassimi | 0549988776 | maha.alqassimi@gmail.com |
| 11 | Fayezah | Salem | 0571234567 | fayez.salem@gmail.com |
| 12 | Zainab | Abdulaziz | 0584567890 | zainab.abdulaziz@gmail.com |

## Service Category Entity & Attributes

Primary key: CategoryID
Foreign key: ManagerID Reference to Employee

| CategoryID | CategoryName | ManagerID |
|---|---|---|
| 1 | Hair Services | 1 |
| 2 | Face Services | 1 |
| 3 | Manicure and Pedicure | 1 |

## Services Category Entity & Attributes

Primary key: ServiceID
Foreign key: CategoryID

| ServiceID | CategoryID | ServiceName | Duration | Price |
|---|---|---|---|---|
| 1 | 1 | Haircut | 00:30:00 | 80.00 |
| 2 | 1 | Hair Coloring | 02:00:00 | 250.00 |
| 3 | 1 | Hair Straightening | 02:00:00 | 200.00 |
| 4 | 1 | Hair Curling | 01:30:00 | 200.00 |
| 5 | 1 | Hair Treatment | 02:00:00 | 300.00 |
| 6 | 1 | Hair Extensions | 02:30:00 | 500.00 |
| 7 | 2 | Facial Cleaning | 01:00:00 | 150.00 |
| 8 | 2 | Makeup | 01:30:00 | 200.00 |
| 9 | 2 | Bridal Makeup | 02:00:00 | 250.00 |
| 10 | 2 | Eyebrow Shaping | 00:30:00 | 50.00 |
| 11 | 2 | Lash Extension | 00:45:00 | 150.00 |
| 12 | 3 | Manicure | 00:30:00 | 80.00 |
| 13 | 3 | Pedicure | 00:45:00 | 90.00 |
| 14 | 3 | Gel Manicure | 00:45:00 | 150.00 |
| 15 | 3 | Nail Art | 01:00:00 | 200.00 |
| 16 | 3 | French Manicure | 00:45:00 | 100.00 |

## Employee Entity & Attributes

Primary key: EmployeeID
Foreign key: CategoryID + ManagerID References to Employee

| EmployeeID | ManagerID | CategoryID | Salary | FirstName | LastName | PhoneNumber | Email |
|---|---|---|---|---|---|---|---|
| 1 | NULL | NULL | 8000.00 | Noha | Saleh | 0551122334 | noha@gmail.com |
| 2 | 1 | 1 | 6000.00 | Lama | Khalid | 0551234567 | lama@gmail.com |
| 3 | 1 | 1 | 5500.80 | Rania | Fahad | 0552345678 | rania@gmail.com |
| 4 | 1 | 1 | 5000.20 | Noura | Saad | 0557891234 | noura@gmail.com |
| 5 | 1 | 1 | 5500.40 | Mona | Ahmed | 0543269536 | maya@gmail.com |
| 6 | 1 | 1 | 5800.30 | Haneen | Salem | 0553214567 | haneen@gmail.com |
| 7 | 1 | 2 | 6500.25 | Dania | Nasser | 0553456789 | dania@gmail.com |
| 8 | 1 | 2 | 5670.00 | Sara | Ali | 0554567890 | sara@gmail.com |
| 9 | 1 | 2 | 5200.80 | Reem | Hadi | 0559876543 | reem@gmail.com |
| 10 | 1 | 2 | 6000.00 | Fatimah | Talal | 0551597536 | fatimah@gmail.com |
| 11 | 1 | 2 | 5100.50 | Nuha | Talal | 0593846552 | nuha@gmail.com |
| 12 | 1 | 2 | 5450.45 | Rahaf | Talal | 0502398334 | rahaf@gmail.com |
| 13 | 1 | 3 | 5300.80 | Huda | Salem | 0555678901 | huda@gmail.com |
| 14 | 1 | 3 | 5400.00 | Amal | Zaid | 0556789012 | amal@gmail.com |
| 15 | 1 | 3 | 5700.60 | Shahad | Omar | 0557418529 | shahed@gmail.com |
| 16 | 1 | 3 | 5400.50 | Asmaa | Zaid | 0583489032 | asmaa@gmail.com |

## Appointment Entity & Attributes

Primary key: AppointmentID
Foreign key: CustomerID+EmployeeID+ServiceID

| Appointment... | CustomerID | EmployeeID | ServiceID | Status | Date | Time | Price |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 12 | Booked | 2025-04-28 | 09:00:00 | 300.00 |
| 2 | 2 | 2 | 5 | Booked | 2025-05-05 | 10:00:00 | 250.00 |
| 3 | 3 | 7 | 3 | Booked | 2025-05-06 | 12:00:00 | 400.00 |
| 4 | 4 | 9 | 1 | Cancelled | 2025-05-06 | 13:00:00 | 100.00 |
| 5 | 5 | 6 | 4 | Cancelled | 2025-05-06 | 14:00:00 | 350.00 |
| 6 | 6 | 4 | 6 | Completed | 2025-05-09 | 15:30:00 | 250.00 |
| 7 | 7 | 3 | 7 | Completed | 2025-05-10 | 16:20:00 | 170.00 |
| 8 | 8 | 13 | 2 | Completed | 2025-05-11 | 17:50:00 | 220.00 |
| 9 | 9 | 10 | 8 | Booked | 2025-05-11 | 18:00:00 | 160.00 |
| 10 | 10 | 5 | 9 | Booked | 2025-05-12 | 19:00:00 | 130.00 |

## Appointment Service Entity (Composite entity)& Attributes

Primary key: None
Foreign key: AppointmentID + ServiceID

| Appointment... | ServiceID |
|---|---|
| 1 | 1 |
| 1 | 2 |
| 4 | 2 |
| 3 | 3 |
| 8 | 3 |
| 10 | 4 |
| 6 | 6 |
| 2 | 7 |
| 7 | 10 |
| 9 | 12 |
| 5 | 16 |

**Payment Entity & Attributes**

Primary key: PaymentID
Foreign key: AppointmentID + CustomerID

| PaymentID | Appointment... | CustomerID | PaymentDate | PaymentMethod | InvoiceNumber |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 2025-04-28 | Bank Transfer | 001 |
| 2 | 2 | 2 | 2025-05-05 | Mada | 002 |
| 3 | 3 | 3 | 2025-05-06 | Bank Transfer | 003 |
| 4 | 4 | 4 | 2025-05-06 | Mada | 004 |
| 5 | 5 | 2 | 2025-05-06 | Mada | 005 |
| 6 | 6 | 3 | 2025-05-09 | Bank Transfer | 006 |
| 7 | 7 | 3 | 2025-05-10 | Mada | 007 |
| 8 | 8 | 1 | 2025-05-11 | Mada | 008 |
| 9 | 9 | 4 | 2025-05-11 | Bank Transfer | 009 |
| 10 | 10 | 4 | 2025-05-12 | Bank Transfer | 010 |

## 2. Logical Design

### Phase I :

# Normalization

**First Normal Form (1NF) :**

Definition:

First Normal Form (1NF) requires that:

- Each column contains atomic (indivisible) values.
- There are no repeating groups or multivalued attributes.

Issue Identified in Our Table:

- In the original Appointment table, some fields (such as ServiceID) included multiple values in one row (e.g., 001, 002, 004), which violates 1NF.
- Also, combining multiple services in one field makes data retrieval and updates more difficult and inconsistent.

Normalization Process:

- We separated the repeating ServiceID groups into a new relation (AppointmentService), where each service linked to a specific appointment is stored in a single row.
- This ensures that every attribute holds only one value per row, satisfying the 1NF requirement.

Result:

- Our data is now atomic and well-structured.
- The relationship between appointments and services is handled through a many-to-many junction table (AppointmentService), which follows the relational model correctly.

| AppointmentID | ServiceID |
|---|---|
| 1 | 1 |
| 1 | 2 |
| 4 | 2 |
| 3 | 3 |
| 8 | 3 |
| 10 | 4 |
| 6 | 6 |
| 2 | 7 |
| 7 | 10 |
| 9 | 12 |
| 5 | 16 |
| NULL | NULL |

| AppointmentID | ServiceID | CustomerID | CustomerEmail |
|---|---|---|---|
| 1 | 001<br>002<br>004 | 1 | sfffse@hghgj |
| 2 | 004<br>007<br>001<br>008 | 2 | effyrgtf@ghjuhi |
| 3 | 009 | 3 | gfjuyfnf@,djfjf |
| 4 | 002<br>003<br>007<br>005 | 4 | Fcgfhgt#567vj@h |

## Second Normal Form (2NF) :

Definition:

A table is in Second Normal Form (2NF) if:

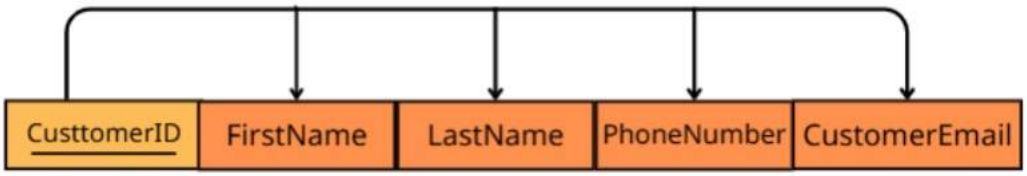
- It is already in First Normal Form (1NF).
- All non-key attributes are fully functionally dependent on the entire primary key (no partial dependency).

Issue Identified in Our Table:

- The Customer table has a single-column primary key (CustomerID), not a composite key.
- All other fields (FirstName, LastName, PhoneNumber, CustomerEmail) depend entirely on CustomerID.
- There is no partial dependency — each attribute is directly related to the primary key.

Result:

- The Customer entity satisfies 2NF.
- No changes were required to bring it into compliance with this form.




## Third Normal Form(3NF) :

Definition:

A table is in Third Normal Form (3NF) if:

- It is already in 2NF.
- There are no transitive dependencies, meaning non-key attributes must depend only on the primary key, and not on another non-key attribute.

Issue Identified :

- In the original Appointment table, the attribute CustomerEmail was included alongside CustomerID.
- However, CustomerEmail does not directly depend on AppointmentID, but rather on CustomerID.
- This creates a transitive dependency:

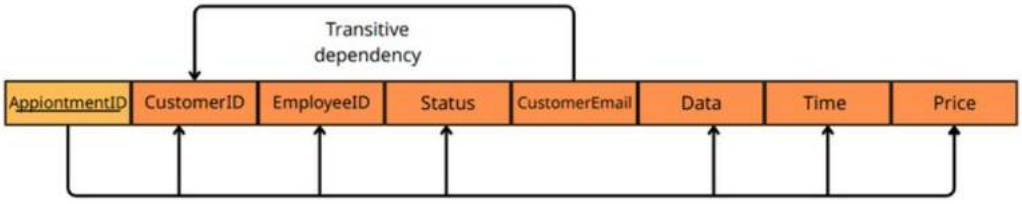
  AppointmentID → CustomerID → CustomerEmail

Normalization process :

- We removed CustomerEmail from the Appointment table.
- The email remains stored in the Customer table, and can be retrieved through a JOIN when needed.

Result:

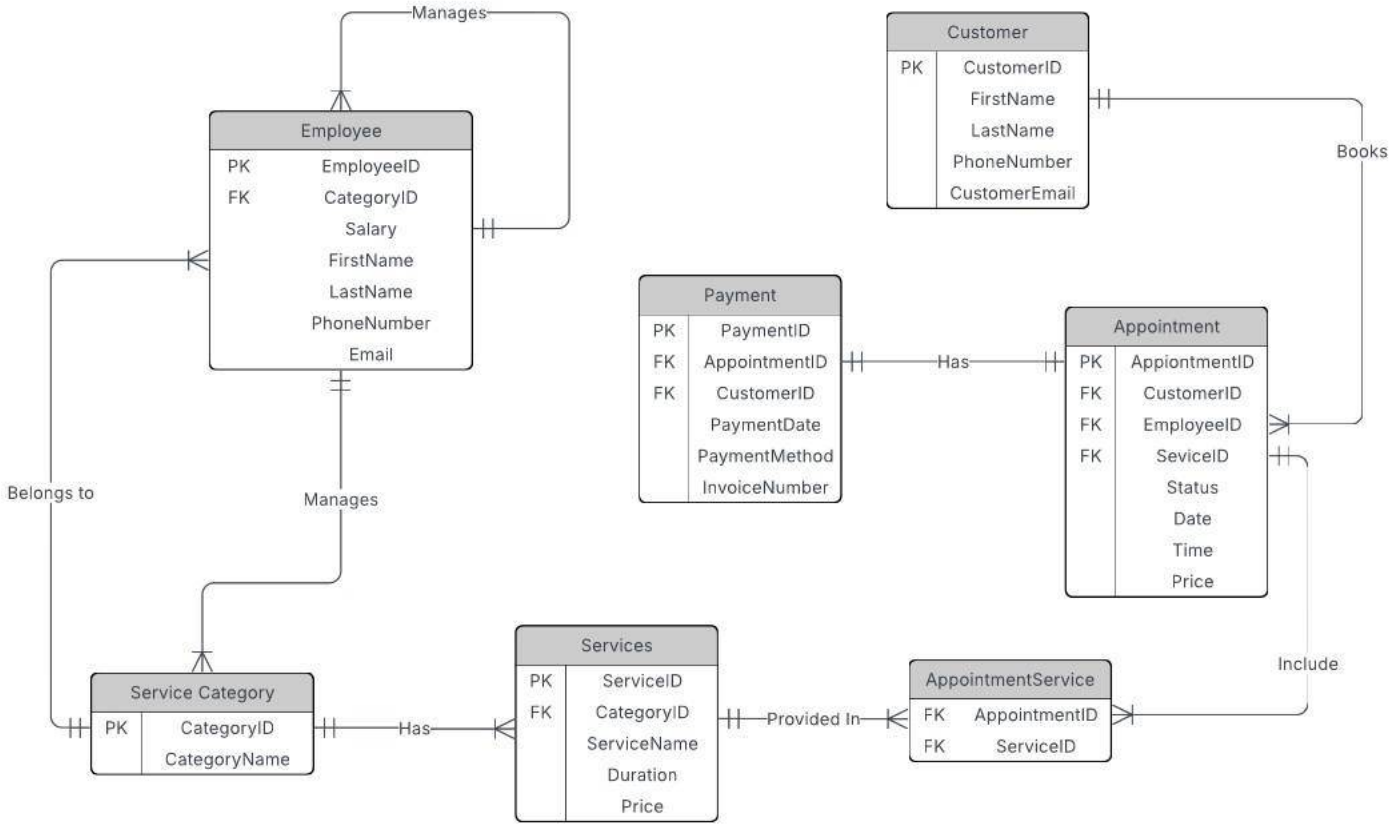
- All non-key attributes in the Appointment table now depend directly on its primary key.
- The table is fully normalized into 3NF.






# Final draft of ERD after Normalization

This final version of the ERD reflects a fully normalized database design that supports the business requirements of the salon reservation system. The relationships are clearly defined, many-to-many associations are properly handled using junction tables (like AppointmentService), and repeating or dependent fields have been resolved to maintain normalization. Unnecessary tables, such as Manager, were removed, and all entities now include essential attributes such as Salary, Status, Duration, and PaymentMethod. The structure is optimized for scalability, integrity, and accurate query handling.



## Phase II :

### Phase II: Logical Design and Implementation

### 1. DDL (Data Definition Language) Implementation

**In this phase, we transformed the conceptual model into actual SQL code. The following tables were created using `CREATE TABLE` statements with proper primary and foreign key constraints:**

- **Customer**
- **ServiceCategory**
- **Employee**
- **Services**
- **Appointment**
- **AppointmentService (junction table for many-to-many relationship)**
- **Payment**

**In addition to defining relationships through foreign keys, constraints such as `CHECK`, `AUTO_INCREMENT`, and `UNIQUE` were used to ensure data integrity.**

**Triggers were also added:**

- **`checktime`: Prevents appointments from being booked outside 09:00–23:00 hours.**
- **`prevent_double_booking`: Ensures no double booking for the same employee at the same time.**

### 2. DML (Data Manipulation Language) Implementation

We populated the database using `INSERT INTO` statements for each table. Here's a summary:

- 12 customers were added.
- 3 service categories (Hair, Face, Nail).
- Multiple employees assigned to each category, including one manager.
- 16 services inserted, classified by their respective categories.
- 10 sample appointments with varying statuses.
- Appointments linked to services via the `AppointmentService` table.
- Payments linked to appointments with appropriate methods (`Mada`, `Bank Transfer`).

### 3. SQL Query Samples

Below are examples of queries executed to test the logic and retrieve meaningful insights from the database:

#### a. List all appointments with customer and employee details:

```sql
SELECT
    a.AppointmentID,
    c.FirstName AS CustomerName,
    e.FirstName AS EmployeeName,
    s.ServiceName,
    a.Status,
    a.Date,
    a.Time,
    a.Price
FROM Appointment a
JOIN Customer c ON a.CustomerID = c.CustomerID
JOIN Employee e ON a.EmployeeID = e.EmployeeID
JOIN Services s ON a.ServiceID = s.ServiceID;
```

#### b. Retrieve completed appointments using `WHERE` condition:

```sql
SELECT AppointmentID, Status, Date, Time
FROM Appointment
WHERE Status = 'Completed';
```

**c. Get appointments between two dates:**

```sql
SELECT *
FROM Appointment
WHERE Date BETWEEN '2025-05-01' AND '2025-05-10';
```

**d. Show total revenue (aggregate function):**

```sql
SELECT SUM(Price) AS TotalRevenue
FROM Appointment
WHERE Status = 'Completed';
```

**e. Count number of services offered in each category:**

```sql
SELECT sc.CategoryName, COUNT(s.ServiceID) AS TotalServices
FROM Services s
JOIN ServiceCategory sc ON s.CategoryID = sc.CategoryID
GROUP BY sc.CategoryName;
```

**f. Find all employees working under a specific category (e.g., Hair Services):**

```sql
SELECT FirstName, LastName
FROM Employee
WHERE CategoryID = (
    SELECT CategoryID
    FROM ServiceCategory
    WHERE CategoryName = 'Hair Services'
);
```

**This phase involved transforming our conceptual design into a fully functioning relational database. Using DDL, we defined tables and constraints; with DML, we populated them; and through queries, we validated data integrity and business logic. The final database is normalized (up to 3NF), scalable, and accurately supports the salon's operations such as bookings, employee management, and payments.**

# Presentation



# Salon Reservation Management System

# Overview

Phase I
Conceptual DB Design

- Business Rules
- Entities & Attributes
- ERD

- Normalization
- Implementation



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# Overview



**Our project is focused on designing and building a Salon Reservation System, aiming to simplify and efficiently manage customer appointments, salon services, and employee assignments. The system will help improve organization and enhance the experience for both the salon staff and clients**

Phase I
Conceptual DB Design

Entities & Attributes

ERD

Normalization

Implementation



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# Requirements Collection and Analysis

## ***Business Rules***

- **Customers can book an appointment only during working hours (9:00 AM – 11:00 PM).**
- **The payment method must be either 'Mada card' or 'Bank Transfer'.**
- **Each appointment must have a status to indicate whether it is booked, cancelled, or completed.**
- **Each service must have a duration that specifies how long it takes to complete.**
- **The manager oversees all service categories and supervises all employees working under these categories**

Phase I
Conceptual DB Design

Business Rules

ERD

Normalization

Implementation



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**Customer**

**Primary key: CustomerID**

**Foreign key: none**

| CustomerID | FirstName | LastName | PhoneNumber | CustomerEmail |
|---|---|---|---|---|
| 1 | Amjad | Mohammed | 0511632582 | amjad.mohammed@gmail.com |
| 2 | Sara | Alhassan | 0552345678 | sara.alhassan@gmail.com |
| 3 | Shaden | Mimesh | 0591863552 | shaden.mimesh@gmail.com |
| 4 | Wafa | Alzahrani | 0531845544 | wafa.alzahrani@gmail.com |
| 5 | Lama | Khalid | 0521023344 | lama.khalid@gmail.com |
| 6 | Hassan | Ali | 0561122334 | hassan.ali@gmail.com |
| 7 | Rania | Fahad | 0509988776 | rania.fahad@gmail.com |
| 8 | Noura | Saad | 0543436789 | noura.saad@gmail.com |
| 9 | Joud | Sami | 0581122333 | joud.sami@gmail.com |
| 10 | Maha | Alqassimi | 0549988776 | maha.alqassimi@gmail.com |
| 11 | Fayezah | Salem | 0571234567 | fayez.salem@gmail.com |
| 12 | Zainab | Abdulaziz | 0584567890 | zainab.abdulaziz@gmail.com |

## Customer Entity & Attributes



## Service Category

**Primary key: CategoryID**

**Foreign key: ManagerID References to Employee**

| CategoryID | CategoryName | ManagerID |
|---|---|---|
| 1 | Hair Services | 1 |
| 2 | Face Services | 1 |
| 3 | Manicure and Pedicure | 1 |

**Service Category Entity & Attributes**



## Services

**Primary key: CustomerID**

**Foreign key: CategoryID**

| ServiceID | CategoryID | ServiceName | Duration | Price |
|---|---|---|---|---|
| 1 | 1 | Haircut | 00:30:00 | 80.00 |
| 2 | 1 | Hair Coloring | 02:00:00 | 250.00 |
| 3 | 1 | Hair Straightening | 02:00:00 | 200.00 |
| 4 | 1 | Hair Curling | 01:30:00 | 200.00 |
| 5 | 1 | Hair Treatment | 02:00:00 | 300.00 |
| 6 | 1 | Hair Extensions | 02:30:00 | 500.00 |
| 7 | 2 | Facial Cleaning | 01:00:00 | 150.00 |
| 8 | 2 | Makeup | 01:30:00 | 200.00 |
| 9 | 2 | Bridal Makeup | 02:00:00 | 250.00 |
| 10 | 2 | Eyebrow Shaping | 00:30:00 | 50.00 |
| 11 | 2 | Lash Extension | 00:45:00 | 150.00 |
| 12 | 3 | Manicure | 00:30:00 | 80.00 |
| 13 | 3 | Pedicure | 00:45:00 | 90.00 |
| 14 | 3 | Gel Manicure | 00:45:00 | 150.00 |
| 15 | 3 | Nail Art | 01:00:00 | 200.00 |
| 16 | 3 | French Manicure | 00:45:00 | 100.00 |

## Services Entity & Attributes



## Employee

**Primary key: EmployeeID**

**Foreign key: CategoryID + ManagerID References to Employee**

| EmployeeID | ManagerID | CategoryID | Salary | FirstName | LastName | PhoneNumber | Email |
|---|---|---|---|---|---|---|---|
| 1 | NULL | NULL | 8000.00 | Noha | Saleh | 0551122334 | noha@gmail.com |
| 2 | 1 | 1 | 6000.00 | Lama | Khalid | 0551234567 | lama@gmail.com |
| 3 | 1 | 1 | 5500.80 | Rania | Fahad | 0552345678 | rania@gmail.com |
| 4 | 1 | 1 | 5000.20 | Noura | Saad | 0557891234 | noura@gmail.com |
| 5 | 1 | 1 | 5500.40 | Mona | Ahmed | 0543269536 | maya@gmail.com |
| 6 | 1 | 1 | 5800.30 | Haneen | Salem | 0553214567 | haneen@gmail.com |
| 7 | 1 | 2 | 6500.25 | Dania | Nasser | 0553456789 | dania@gmail.com |
| 8 | 1 | 2 | 5670.00 | Sara | Ali | 0554567890 | sara@gmail.com |
| 9 | 1 | 2 | 5200.80 | Reem | Hadi | 0559876543 | reem@gmail.com |
| 10 | 1 | 2 | 6000.00 | Fatimah | Talal | 0551597536 | fatimah@gmail.com |
| 11 | 1 | 2 | 5100.50 | Nuha | Talal | 0593846552 | nuha@gmail.com |
| 12 | 1 | 2 | 5450.45 | Rahaf | Talal | 0502398334 | rahaf@gmail.com |
| 13 | 1 | 3 | 5300.80 | Huda | Salem | 0555678901 | huda@gmail.com |
| 14 | 1 | 3 | 5400.00 | Amal | Zaid | 0556789012 | amal@gmail.com |
| 15 | 1 | 3 | 5700.60 | Shahad | Omar | 0557418529 | shahed@gmail.com |
| 16 | 1 | 3 | 5400.50 | Asmaa | Zaid | 0583489032 | asmaa@gmail.com |

## Employee Entity & Attributes



## Appointment

**Primary key: AppointmentID**

**Foreign key: CustomerID + EmployeeID + ServiceID**

| Appointment... | CustomerID | EmployeeID | ServiceID | Status | Date | Time | Price |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 12 | Booked | 2025-04-28 | 09:00:00 | 300.00 |
| 2 | 2 | 2 | 5 | Booked | 2025-05-05 | 10:00:00 | 250.00 |
| 3 | 3 | 7 | 3 | Booked | 2025-05-06 | 12:00:00 | 400.00 |
| 4 | 4 | 9 | 1 | Cancelled | 2025-05-06 | 13:00:00 | 100.00 |
| 5 | 5 | 6 | 4 | Cancelled | 2025-05-06 | 14:00:00 | 350.00 |
| 6 | 6 | 4 | 6 | Completed | 2025-05-09 | 15:30:00 | 250.00 |
| 7 | 7 | 3 | 7 | Completed | 2025-05-10 | 16:20:00 | 170.00 |
| 8 | 8 | 13 | 2 | Completed | 2025-05-11 | 17:50:00 | 220.00 |
| 9 | 9 | 10 | 8 | Booked | 2025-05-11 | 18:00:00 | 160.00 |
| 10 | 10 | 5 | 9 | Booked | 2025-05-12 | 19:00:00 | 130.00 |

## Appointment Entity & Attributes



## Appointment Service

**Primary key: none**

**Foreign key: AppointmentID + ServiceID**

| | Appointment... | ServiceID |
|---|---|---|
| | 1 | 1 |
| | 1 | 2 |
| | 4 | 2 |
| | 3 | 3 |
| | 8 | 3 |
| | 10 | 4 |
| | 6 | 6 |
| | 2 | 7 |
| | 7 | 10 |
| | 9 | 12 |
| | 5 | 16 |

**Appointment Sevices Entity (Composite Entity) & Attributes**



## Payment

**Primary key: PaymentID**

**Foreign key: AppointmentID + CustomerID**

| PaymentID | Appointment... | CustomerID | PaymentDate | PaymentMethod | InvoiceNumber |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 2025-04-28 | Bank Transfer | 001 |
| 2 | 2 | 2 | 2025-05-05 | Mada | 002 |
| 3 | 3 | 3 | 2025-05-06 | Bank Transfer | 003 |
| 4 | 4 | 4 | 2025-05-06 | Mada | 004 |
| 5 | 5 | 2 | 2025-05-06 | Mada | 005 |
| 6 | 6 | 3 | 2025-05-09 | Bank Transfer | 006 |
| 7 | 7 | 3 | 2025-05-10 | Mada | 007 |
| 8 | 8 | 1 | 2025-05-11 | Mada | 008 |
| 9 | 9 | 4 | 2025-05-11 | Bank Transfer | 009 |
| 10 | 10 | 4 | 2025-05-12 | Bank Transfer | 010 |

**Payment Entity & Attributes**



Phase I
Conceptual DB Design

Business Rules

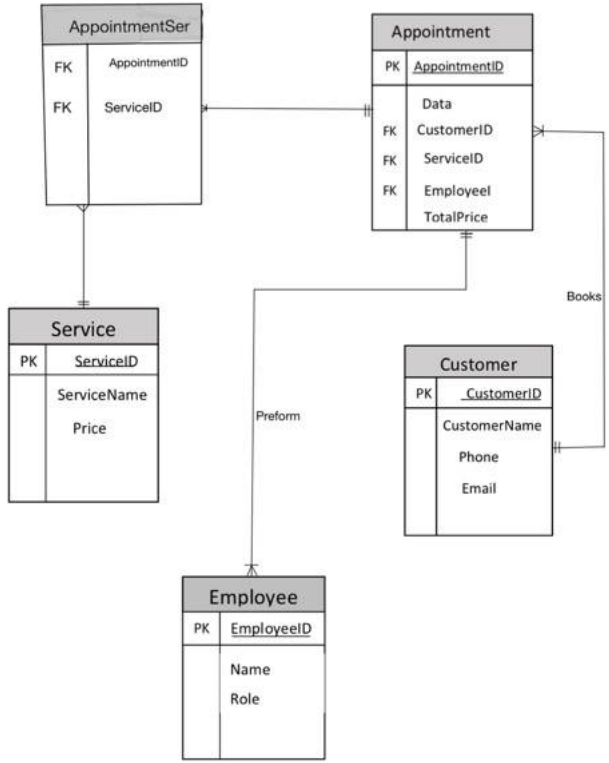
Entities & Attributes

Normalization

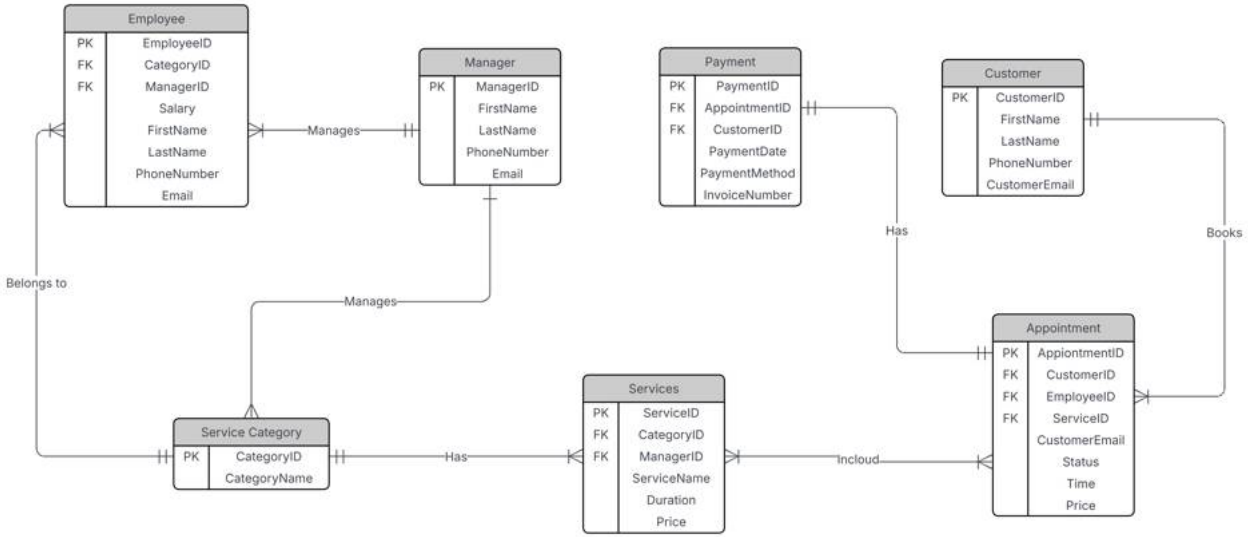
Implementation



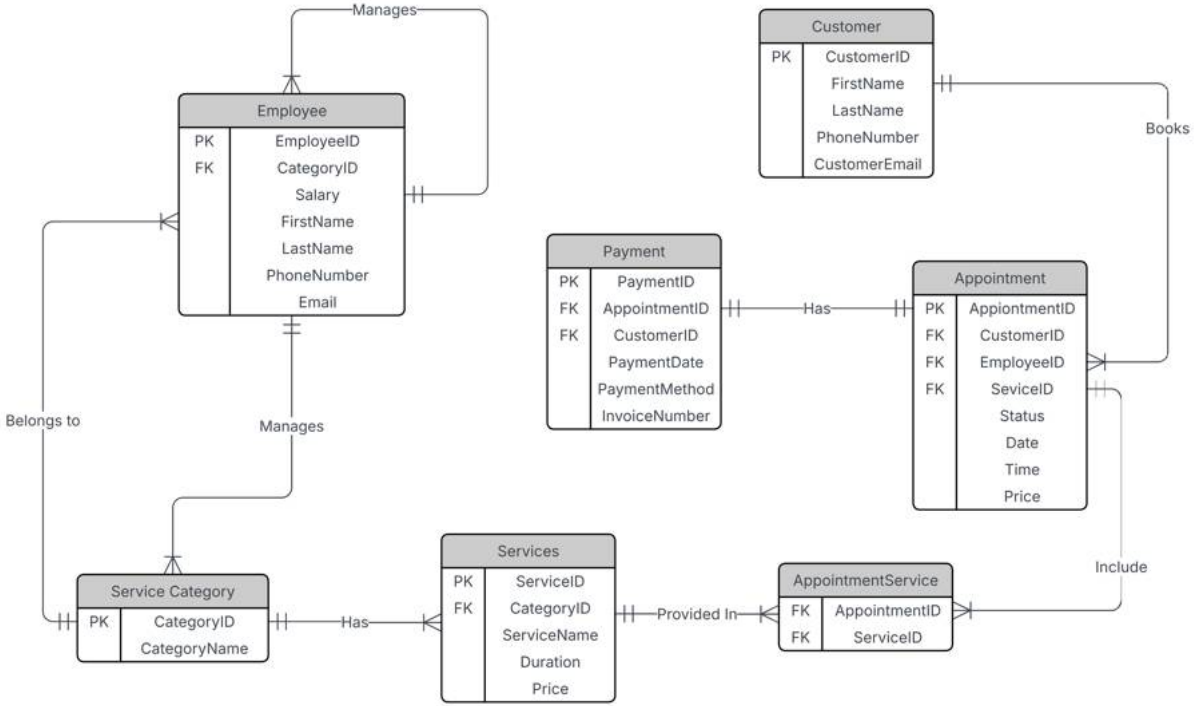
WASS Beauty


AppointmentSer
FK
AppointmentID
FK
ServiceID
Appointment
PK
AppointmentID
Data
FK
CustomerID
FK
ServiceID
FK
EmployeeI
TotalPrice
Books
Service
PK
ServiceID
ServiceName
Price
Preform
Customer
PK
CustomerID
CustomerName
Phone
Email
Employee
PK
EmployeeID
Name
Role

WASS Beauty


Employee
PK EmployeeID
FK CategoryID
FK ManagerID
Salary
FirstName
LastName
PhoneNumber
Email
Manages
Manager
PK ManagerID
FirstName
LastName
PhoneNumber
Email
Payment
PK PaymentID
FK AppointmentID
FK CustomerID
PaymentDate
PaymentMethod
InvoiceNumber
Customer
PK CustomerID
FirstName
LastName
PhoneNumber
CustomerEmail
Has
Books
Belongs to
Manages
Appointment
PK AppiontmentID
FK CustomerID
FK EmployeeID
FK ServiceID
CustomerEmail
Status
Time
Price
Service Category
PK CategoryID
CategoryName
Has
Services
PK ServiceID
FK CategoryID
FK ManagerID
ServiceName
Duration
Price
Incloud

WASS Beauty


Manages
Employee
PK EmployeeID
FK CategoryID
Salary
FirstName
LastName
PhoneNumber
Email
Customer
PK CustomerID
FirstName
LastName
PhoneNumber
CustomerEmail
Books
Payment
PK PaymentID
FK AppointmentID
FK CustomerID
PaymentDate
PaymentMethod
InvoiceNumber
Has
Appointment
PK AppiontmentID
FK CustomerID
FK EmployeeID
FK SeviceID
Status
Date
Time
Price
Belongs to
Manages
Service Category
PK CategoryID
CategoryName
Has
Services
PK ServiceID
FK CategoryID
ServiceName
Duration
Price
Provided In
AppointmentService
FK AppointmentID
FK ServiceID
Include



WASS Beauty

Phase I
Conceptual DB Design

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ERD

Implementation



WASS Beauty

# NORMALIZATION

## FIRST NORMAL FORM(1NF)

A table is in 1NF if each column do not contain Multi-valurd groups, every field holds only on value

in our table: there are repeating groups

| | AppointmentID | ServiceID |
|---|---|---|
| ▶ | 1 | 1 |
| | 1 | 2 |
| | 4 | 2 |
| | 3 | 3 |
| | 8 | 3 |
| | 10 | 4 |
| | 6 | 6 |
| | 2 | 7 |
| | 7 | 10 |
| | 9 | 12 |
| | 5 | 16 |
| * | NULL | NULL |

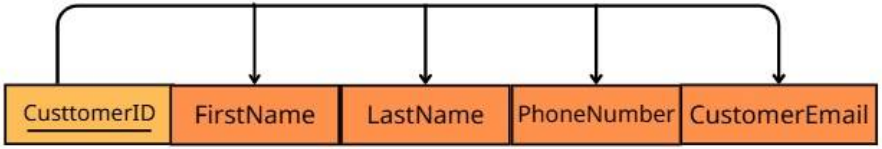
| AppointmentID | ServiceID | CustomerID | CustomerEmail |
|---|---|---|---|
| 1 | 001<br>002<br>004 | 1 | sfffse@hghgj |
| 2 | 004<br>007<br>001<br>008 | 2 | effyrgtf@ghjuhi |
| 3 | 009 | 3 | gfjuyfnf@,djfjf |
| 4 | 002<br>003<br>007<br>005 | 4 | Fcgfhgt#567vj@h |

NORMALIZATION

# SECOND NORMAL FORM(2NF)

it must be in 1NF, and all attributes depend on primary key (no partial dependency)

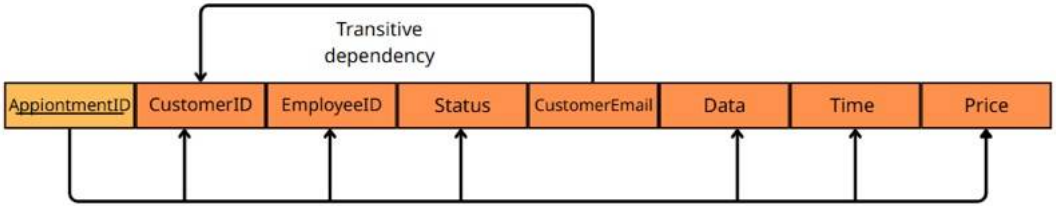
in our table: no Partial dependencies



# NORMALIZATION

## THIRD NORMAL FORM(3NF)

it must be in 2NF, and no transitive dependencies in our table: we noticed one issue in appointment table




| Appointment | |
|---|---|
| PK | AppiontmentID |
| FK | CustomerID |
| FK | EmployeeID |
| | Status |
| | CustomerEmail |
| | Date |
| | Time |
| | Price |

Phase I
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WASS Beauty

# DDL, DML, and Query Samples

## DDL Implementation

Create table

```
CREATE TABLE Appointment (
    AppointmentID INT AUTO_INCREMENT PRIMARY KEY,
    CustomerID INT NOT NULL,
    EmployeeID INT NOT NULL,
    ServiceID INT NOT NULL,
    Status VARCHAR(50) NOT NULL,
    CHECK (Status IN ('Booked', 'Cancelled', 'Completed')),
    Date DATE NOT NULL,
    Time TIME NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID),
    FOREIGN KEY (ServiceID) REFERENCES Services(ServiceID)
);
```

CHECK (constraint)

```sql
CREATE TABLE Payment (
  PaymentID INT AUTO_INCREMENT PRIMARY KEY,
  AppointmentID INT UNIQUE,
  CustomerID INT NOT NULL,
  PaymentDate DATE NOT NULL,
  PaymentMethod  Varchar (50) NOT NULL,
  check (PaymentMethod in('Mada', 'Bank Transfer')),
  InvoiceNumber VARCHAR(50) NOT NULL,
  FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID),
  FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
);
```

```sql
CREATE TABLE Appointment (
  AppointmentID INT AUTO_INCREMENT PRIMARY KEY,
  CustomerID INT NOT NULL,
  EmployeeID INT NOT NULL,
  ServiceID INT NOT NULL,
  Status VARCHAR(50) NOT NULL,
  CHECK (Status IN ('Booked', 'Cancelled', 'Completed')),
  Date DATE NOT NULL,
  Time TIME NOT NULL,
  Price DECIMAL(10,2) NOT NULL,
  FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
  FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID)
  FOREIGN KEY (ServiceID) REFERENCES Services(ServiceID)
);
```

# Triggers

```
USE WASSBeautySalon;
CREATE TRIGGER checktime
BEFORE INSERT ON Appointment
FOR EACH ROW
BEGIN
    IF NEW.Time < '09:00:00' OR NEW.Time > '23:00:00' THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Appointment time must be between 09:00 and 23:00.';
    END IF;
END//
DELIMITER ;
```



# Triggers

```
CREATE TRIGGER prevent_double_booking
BEFORE INSERT ON Appointment
FOR EACH ROW
BEGIN
    IF EXISTS (
        SELECT 1
        FROM Appointment
        WHERE employeeID = NEW.employeeID
          AND Date = NEW.Date
          AND Time = NEW.Time
    ) THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'This appointment is already booked for this service at this time.';
    END IF;
END//
DELIMITER ;
```



add

```
alter table appointmentservice
add ServiceName varchar(50);
```

| AppointmentID | ServiceID | ServiceName |
|---|---|---|
| 1 | 1 | NULL |
| 1 | 2 | NULL |
| 2 | 7 | NULL |
| 3 | 3 | NULL |
| 4 | 2 | NULL |
| 5 | 16 | NULL |
| 6 | 6 | NULL |
| 7 | 10 | NULL |
| 8 | 3 | NULL |

rename

```
1    alter table employee
2    rename to Employees;
```

modify

```
1 •  alter table appointmentservice
2    modify ServiceName varchar(60);
```

drop

```
1 •  alter table appointmentservice
2    drop column ServiceName;
```

# DDL, DML, and Query Samples

## DML Implementation

insert

```
INSERT INTO Employee (ManagerID, CategoryID, FirstName, LastName, PhoneNumber, Email, Salary) VALUES
(1, 3, 'Soha', 'Fahad', '0542668368', 'Soha@gmail.com', 5600.00);
```

# DDL, DML, and Query Samples

## Query Samples

```sql
SELECT *
FROM Services
WHERE Price > 150 AND Duration > '01:00:00';

SELECT *
FROM Appointment
WHERE Status = 'Booked' OR Status = 'Completed';

SELECT Customer.FirstName, Customer.LastName, Appointment.Date, Appointment.Time, Appointment.Status
FROM Appointment
INNER JOIN Customer ON Appointment.CustomerID = Customer.CustomerID;


SELECT Status, COUNT(*) AS TotalAppointments
FROM Appointment
GROUP BY Status;
```

# THANK YOU

